Land 14: Nephra Springs

Urinary system, fluids, and acid base · OpenStax Ch. 25 and 26

Key ideas

Urine and kidney gross anatomy

- Normal urine is clear to amber, slightly acidic (about pH 6, range 4.5 to 8), and holds urea, creatinine, uric acid and ions; glucose, protein and blood are abnormal.
- Daily output is roughly 1 to 2 L; oliguria is below about 400 mL per day, anuria below about 50 mL, polyuria above about 2.5 L.
- Kidneys are retroperitoneal, with the right a bit lower because of the liver; they are wrapped by the renal capsule, adipose capsule and renal fascia.
- The outer cortex surrounds the inner medulla, which forms renal pyramids separated by renal columns of cortex.
- Urine path: papilla, minor calyx, major calyx, renal pelvis, ureter.

Renal blood supply and nephron structure

- The kidneys receive about 25 percent of cardiac output through the renal arteries.
- Arterial path: renal, segmental, interlobar, arcuate, cortical radiate, afferent arteriole, glomerulus, efferent arteriole, then peritubular capillaries or vasa recta.
- The renal corpuscle (glomerulus plus glomerular capsule) filters; the tubule runs PCT, loop of Henle, DCT, then collecting duct.
- Cortical nephrons (about 85 percent) have short loops; juxtamedullary nephrons have long loops and vasa recta that concentrate urine.
- PCT cells carry a microvillus brush border; collecting ducts have principal cells (Na+ and water) and intercalated cells (acid base).

Glomerular filtration and GFR

- The filtration membrane is fenestrated endothelium, basement membrane and podocyte filtration slits; blood cells and most proteins stay in the blood.
- NFP equals glomerular blood hydrostatic pressure minus the sum of capsular hydrostatic and blood colloid osmotic pressures; typically 55 minus (15 plus 30) gives 10 mm Hg.
- Normal GFR is about 125 mL per minute in men and a bit lower in women, roughly 180 L of filtrate per day.
- The afferent arteriole is wider than the efferent arteriole, keeping glomerular pressure high.
- Clearance equals urine concentration times urine flow divided by plasma concentration; inulin clearance equals GFR, and creatinine clearance estimates it.

Regulating GFR: autoregulation and the JGA

- Autoregulation holds GFR fairly steady across a wide range of mean arterial pressures (about 80 to 180 mm Hg).
- Myogenic mechanism: a stretched afferent arteriole contracts; a relaxed one dilates.
- Tubuloglomerular feedback: macula densa cells sense NaCl in the tubular fluid and adjust afferent arteriole diameter.
- The juxtaglomerular apparatus includes macula densa cells, renin secreting juxtaglomerular cells of the afferent arteriole, and extraglomerular mesangial cells.
- Strong sympathetic stimulation constricts afferent arterioles, lowering GFR and urine output to conserve volume.

Reabsorption, secretion and concentrating urine

- The PCT reabsorbs about 65 percent of Na+ and water and nearly all glucose and amino acids; glucose spills into urine only when the transport maximum is exceeded.
- The descending limb is permeable to water; the thick ascending limb actively pumps out Na+, K+ and chloride but is impermeable to water, so its fluid leaves hypotonic.
- The countercurrent multiplier builds a medullary gradient that rises toward the papilla; urea recycling adds to it, and the vasa recta preserve it as a countercurrent exchanger.
- Secretion moves H+, K+, creatinine and many drugs from peritubular blood into the tubule.
- Overall, more than 99 percent of filtrate is reabsorbed.

Hormones: ADH, aldosterone, ANP and RAAS

- ADH from the posterior pituitary inserts aquaporin 2 channels in collecting duct cells, producing small volumes of concentrated urine.
- Aldosterone from the adrenal cortex increases Na+ reabsorption and K+ secretion in the late DCT and collecting duct; high K+ and angiotensin II stimulate it.
- RAAS: low pressure triggers renin, which converts angiotensinogen to angiotensin I; ACE (mainly in the lungs) forms angiotensin II.
- Angiotensin II constricts arterioles, stimulates aldosterone, ADH and thirst, and raises blood pressure.
- ANP from stretched atria raises GFR and inhibits renin, aldosterone and ADH, increasing Na+ and water loss.

Ureters, bladder, urethra and micturition

- Ureters move urine by peristalsis and enter the bladder at an angle, so a full bladder squeezes them shut and limits backflow.
- The bladder has a transitional epithelium, folds called rugae, the detrusor muscle, and the trigone outlined by two ureteral openings and the urethral opening.
- The internal urethral sphincter is involuntary smooth muscle; the external urethral sphincter is voluntary skeletal muscle.
- The female urethra is about 4 cm and the male about 20 cm, so UTIs are more common in females.
- Micturition reflex: stretch receptors signal the sacral cord, parasympathetic fibers contract the detrusor, and brain centers allow voluntary control of the external sphincter.

Body fluids, water balance and electrolytes

- About 60 percent of an adult male's body mass is water; two thirds is ICF and one third ECF (mostly interstitial fluid, about 20 percent of ECF is plasma).
- Thirst and ADH respond to rising osmolarity sensed by hypothalamic osmoreceptors and to falling blood volume; urine is the largest regulated water loss.
- Na+ is the main ECF cation and chloride the main ECF anion; K+ is the main ICF cation and phosphate the main ICF anion.
- High K+ is dangerous for the heart; aldosterone increases K+ excretion.
- PTH raises blood Ca2+ and increases phosphate excretion; calcitonin lowers blood Ca2+; low Ca2+ increases nerve and muscle excitability.

Acid base balance: buffers, regulation and disorders

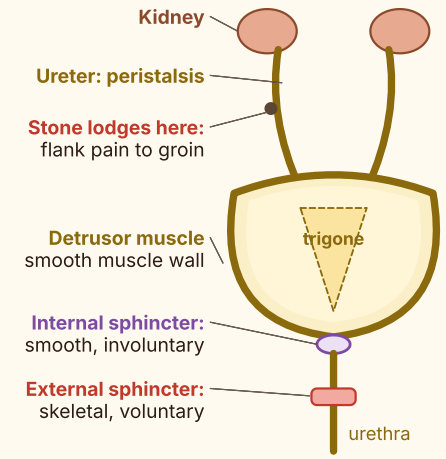
- Normal arterial pH is 7.35 to 7.45 and normal PCO2 about 35 to 45 mm Hg.
- Chemical buffers act in seconds: bicarbonate (blood), phosphate (cells and urine) and proteins including hemoglobin.
- The lungs adjust CO2 within minutes; the kidneys act over hours to days by secreting H+, reclaiming and making bicarbonate, and excreting acid with phosphate and ammonia.
- Respiratory disorders come from CO2: hypoventilation causes acidosis, hyperventilation alkalosis; the kidneys compensate.
- Metabolic disorders come from bicarbonate or fixed acids: diarrhea and ketoacidosis cause acidosis, vomiting causes alkalosis; breathing compensates.

Villager wisdom

- Here in Nephra Springs we say the nephron is a sieve then a sorter: the glomerulus filters nearly everything small, and the tubules take back whatever the village still needs.
- Remember PCT as Pretty much Collects Things. About two thirds of the salt and water, and every bit of the glucose, is gathered back there before the loop even begins.
- ADH puts doors in the dam. It slips aquaporins into the collecting ducts so spring water flows back to the blood and the urine runs dark and scant.
- Aldosterone is a salty old miser, traveler: it hoards the sodium and pays out the potassium.
- Travel to ROME when reading blood gases: Respiratory Opposite, Metabolic Equal. In respiratory troubles pH and CO2 move opposite ways; in metabolic ones pH and bicarbonate move together.

Diagrams

Urinary tract and micturition



Micturition reflex

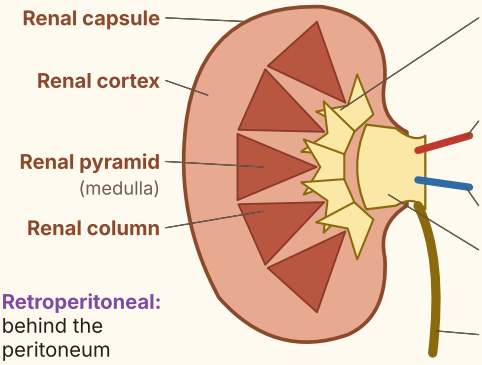
bladder stretch receptors signal the sacral cord; parasympathetic nerves contract the detrusor and open the internal sphincter; you relax the external sphincter to void

Female urethra about 4 cm, near the anus: more UTIs

Normal urine: pH about 6; water, urea, creatinine, ions

Ureters move urine by peristalsis; during micturition parasympathetic nerves contract the detrusor muscle, while the external urethral sphincter is skeletal muscle under voluntary control. The short female urethra opening near the anus makes urinary tract infections more common in females.

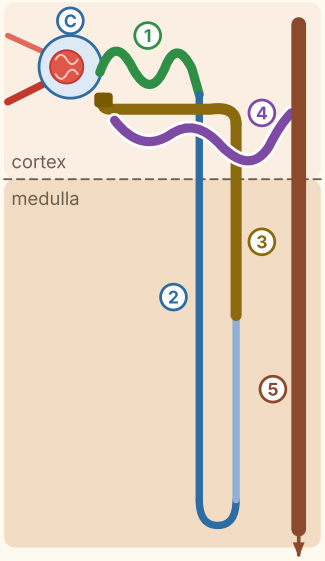
Kidney anatomy and blood supply



Blood path: renal artery, segmental, interlobar, arcuate, cortical radiate, afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries and vasa recta, veins

Urine drips from each renal papilla into a minor calyx, then a major calyx, the renal pelvis and the ureter. The kidneys are retroperitoneal, and each glomerulus is fed by an afferent arteriole and drained by an efferent arteriole.

The nephron: what happens where



C Renal corpuscle
glomerulus in glomerular (Bowman) capsule: filtration

1 Proximal convoluted tubule
reabsorbs about 65 percent of water and Na+, all glucose and amino acids; secretes H+ and drugs

2 Descending limb
water leaves; no NaCl pumped

3 Thick ascending limb
pumps out NaCl, keeps water: fluid leaves hypotonic; loop diuretics block this pump

4 Distal convoluted tubule
Na+, Ca2+ (PTH) reabsorbed; macula densa senses NaCl

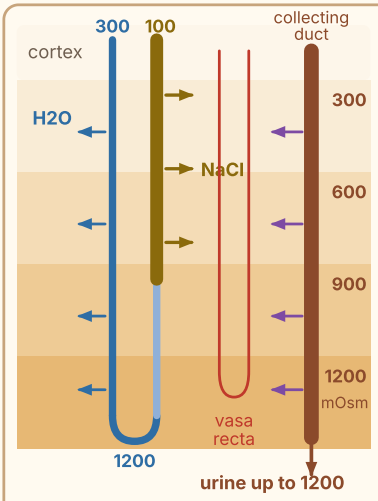
5 Collecting duct
ADH: aquaporins, water out; aldosterone: Na+ in, K+ out

180 L filtered a day, about 1.8 L of urine: 99 percent reabsorbed

Juxtamedullary nephrons have long loops reaching deep into the medulla

Filtrate flows from the glomerular capsule through the PCT, loop of Henle and DCT into the collecting duct. The PCT reabsorbs most water and Na+ and all the glucose, while hormones fine tune the end: ADH for water and aldosterone for Na+ and K+.

Concentrating urine: countercurrent and ADH



Countercurrent multiplier
descending limb loses water; thick ascending limb pumps out NaCl but keeps water, so the medulla gets salty

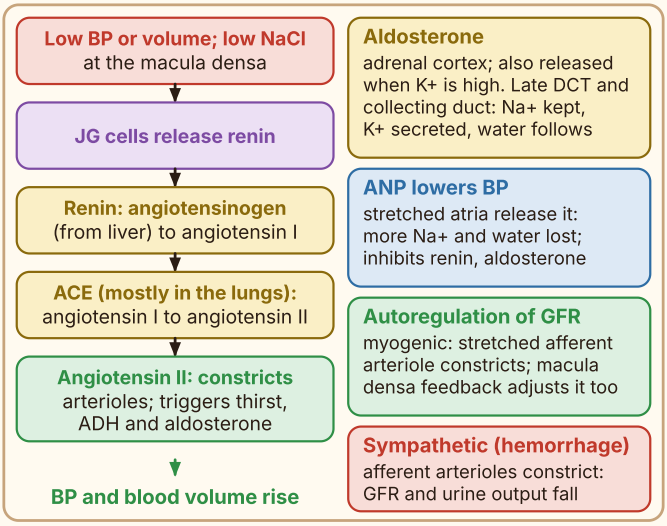
Vasa recta
countercurrent exchanger; keeps the gradient

ADH
salty blood: osmoreceptors in the hypothalamus cause thirst and ADH release; ADH puts aquaporins in the collecting duct: little, concentrated urine

No ADH (diabetes insipidus): many liters of dilute urine

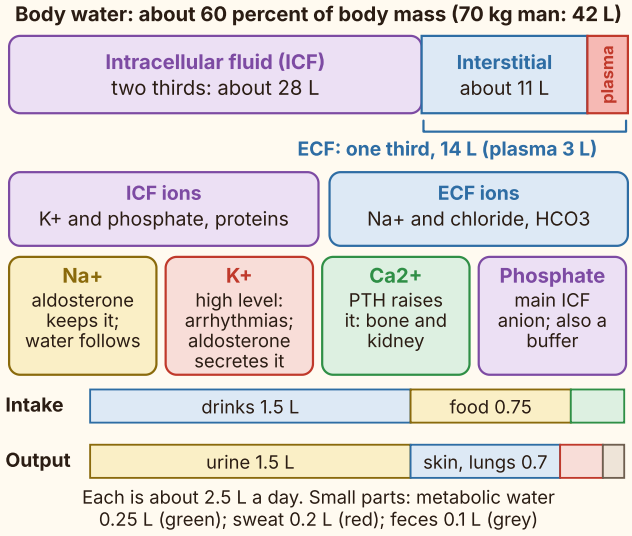
The loop of Henle is a countercurrent multiplier that builds a salty medulla, and the vasa recta preserve it as a countercurrent exchanger. ADH inserts aquaporins so collecting duct water follows that gradient, giving a small volume of concentrated urine; without ADH urine stays dilute.

Hormones that adjust the kidney



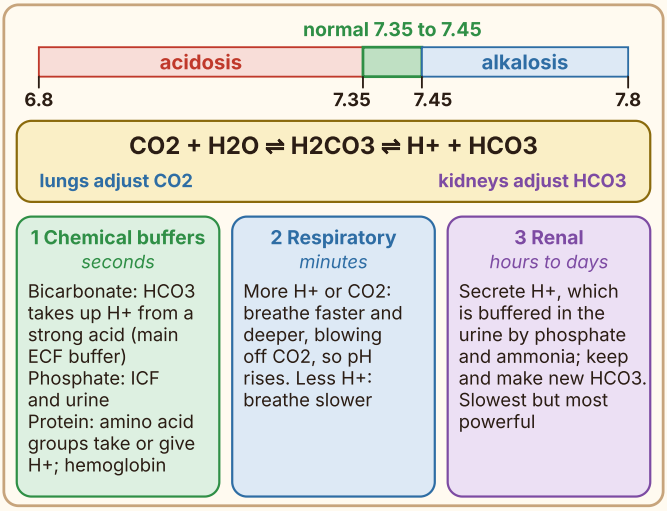
Low blood pressure or low NaCl at the macula densa triggers renin; ACE turns angiotensin I into angiotensin II, which constricts vessels and releases aldosterone so Na+ and water are kept and K+ is secreted. ANP from stretched atria does the opposite and lowers blood pressure.

Body fluids and electrolytes



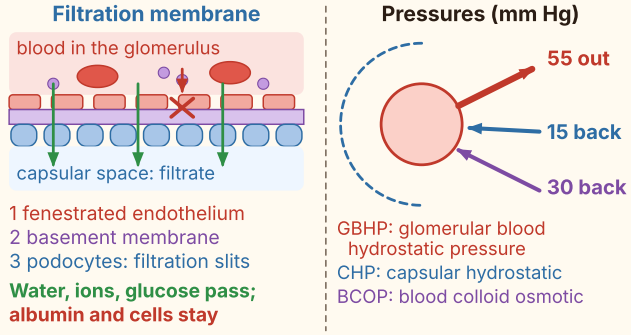
About two thirds of body water is intracellular, where K+ and phosphate dominate; extracellular fluid is rich in Na+ and chloride. Urine is the largest daily route of water loss, and thirst is driven by osmoreceptors in the hypothalamus.

Acid base balance: buffers and compensation



Chemical buffers act in seconds, the lungs adjust CO2 within minutes, and the kidneys correct pH over hours to days by secreting H+ and reclaiming or making bicarbonate. In the bicarbonate buffer, bicarbonate ion takes up the H+ from a strong acid.

Glomerular filtration



Net filtration pressure is glomerular blood hydrostatic pressure minus the capsular hydrostatic and blood colloid osmotic pressures, about 10 mm Hg. Water and small solutes cross the three layer membrane but proteins such as albumin and blood cells do not.

Reading acid base disorders

Acidosis: pH below 7.35	Alkalosis: pH above 7.45
Respiratory acidosis **PCO2 ↑ (above 45)** hypoventilation: opioids, COPD, airway blockage *Compensation: kidneys raise HCO3*	**Respiratory alkalosis** **PCO2 ↓ (below 35)** hyperventilation: panic, high altitude *Compensation: kidneys lower HCO3*
Metabolic acidosis **HCO3 ↓ (below 22)** ketoacidosis, diarrhea, kidney failure *Compensation: lungs lower PCO2*	**Metabolic alkalosis** **HCO3 ↑ (above 26)** vomiting stomach acid, too much antacid *Compensation: lungs raise PCO2*

Steps

1 pH: acidosis or alkalosis?
2 Which value matches? PCO2 (35 to 45): respiratory; HCO3 (22 to 26 mEq/L): metabolic
3 Other value moving the same way: compensating; pH back in range: fully compensated

Respiratory disorders start with PCO2 and metabolic disorders start with bicarbonate; the other value moving in the same direction is compensation. A normal pH with both values abnormal means full compensation.

Self check

1. **Osmoreceptors that sense rising blood osmolarity and trigger thirst are located in the:**
    A. Renal cortex
    B. Medulla oblongata
    C. Hypothalamus
    D. Adrenal cortex
2. **A patient with long term COPD has pH 7.37, PCO2 60 mm Hg, and bicarbonate 34 mEq/L. Best interpretation?**
    A. Metabolic alkalosis, uncompensated
    B. Normal acid base status
    C. Metabolic acidosis, respiratory compensation
    D. Respiratory acidosis, fully compensated
3. **Why are urinary tract infections more common in females than in males?**
    A. Females lack an internal urethral sphincter
    B. The urethra is short and opens near the anus
    C. Female urine is strongly alkaline
    D. Females have only one ureter
4. **Which urethral sphincter is skeletal muscle under voluntary control?**
    A. Internal urethral sphincter
    B. External urethral sphincter
    C. Detrusor muscle
    D. Trigone
5. **A hiker sweats away 2 L of water on a hot day and drinks nothing. Which response is expected?**
    A. More ADH and a small volume of concentrated urine
    B. Less ADH and a large volume of dilute urine
    C. More ANP and more Na+ excretion
    D. Less renin and less aldosterone
6. **The thick ascending limb pumps out NaCl but is impermeable to water. Fluid leaving it is therefore:**
    A. Hypertonic to plasma
    B. Hypotonic to plasma
    C. Isotonic to plasma
    D. Free of all Na+
7. **About what fraction of total body water is intracellular fluid?**
    A. About one third
    B. About two thirds
    C. About one half
    D. About one tenth

8. **Phosphate is the most abundant anion in which fluid?**
    A. Blood plasma
    B. Interstitial fluid
    C. Cerebrospinal fluid
    D. Intracellular fluid
9. **What is the normal range of arterial blood pH?**
    A. 6.8 to 7.0
    B. 7.35 to 7.45
    C. 7.0 to 7.2
    D. 7.45 to 7.65
10. **Aldosterone acts on the late distal tubule and collecting duct to:**
    A. Reabsorb Na+ and secrete K+
    B. Reabsorb K+ and secrete Na+
    C. Insert aquaporins only
    D. Secrete renin
11. **In the renin angiotensin aldosterone system, angiotensin converting enzyme (ACE) converts:**
    A. Angiotensinogen to angiotensin I
    B. Angiotensin II to aldosterone
    C. Angiotensin I to angiotensin II
    D. Renin to angiotensinogen
12. **How do the kidneys correct an acidosis over hours to days?**
    A. Excrete more bicarbonate in urine
    B. Reduce ammonia production
    C. Increase CO_2 exhalation
    D. Secrete H+ and reabsorb or make bicarbonate

Answer key

1. **C.** Hypothalamus. Hypothalamic osmoreceptors detect concentrated plasma, stimulating thirst and ADH release from the posterior pituitary. The medulla oblongata is tempting since it controls breathing and heart rate, but it is not the thirst center.
2. **D.** Respiratory acidosis, fully compensated. Retained CO_2 is the primary problem, and over days the kidneys raised bicarbonate enough to bring pH back into the normal range. Normal status is tempting because pH is normal, but PCO_2 and bicarbonate are both clearly abnormal.
3. **B.** The urethra is short and opens near the anus. The female urethra is only about 4 cm long and its opening is close to the anus, so intestinal bacteria reach the bladder easily. The sphincter option is tempting, but the key difference is the short length and location of the urethra.
4. **B.** External urethral sphincter. The external urethral sphincter is skeletal muscle of the pelvic floor that we learn to control. The internal sphincter is tempting, but it is smooth muscle at the bladder neck and is involuntary.
5. **A.** More ADH and a small volume of concentrated urine. Rising plasma osmolarity and falling volume trigger thirst and ADH, which inserts aquaporins so the collecting ducts save water. ANP is tempting as a volume hormone, but it is released when the atria are stretched by too much volume, not too little.

6. **B.** Hypotonic to plasma. Removing salt while keeping water in the lumen dilutes the fluid, so it enters the DCT hypotonic. Hypertonic is tempting because the fluid is concentrated at the hairpin turn, but the ascending limb strips solute out on the way up.
7. **B.** About two thirds. Roughly two thirds of body water is inside cells (ICF). One third is tempting, but that is the extracellular fluid, which is mostly interstitial fluid plus plasma.
8. **D.** Intracellular fluid. Phosphate is concentrated inside cells, where it is built into ATP, nucleic acids and phospholipids and also buffers pH. Plasma is tempting, but chloride is the dominant anion in plasma and other extracellular fluids.
9. **B.** 7.35 to 7.45. Arterial blood is normally slightly alkaline, between 7.35 and 7.45. Values below 7.35 are acidosis and above 7.45 are alkalosis, so 7.45 to 7.65 lies entirely in the alkalotic zone.
10. **A.** Reabsorb Na+ and secrete K+. Aldosterone increases Na+ reabsorption (water follows) while increasing K+ secretion into the urine. The reversed option is the classic trap; remember aldosterone keeps salt and spends potassium.
11. **C.** Angiotensin I to angiotensin II. ACE, found mainly in the lungs, turns angiotensin I into the active angiotensin II. Angiotensinogen to angiotensin I is tempting, but that earlier step is catalyzed by renin from the JG cells.
12. **D.** Secrete H+ and reabsorb or make bicarbonate. Tubule cells secrete H+ into the filtrate while reclaiming filtered bicarbonate and generating new bicarbonate for the blood. Increasing CO_2 exhalation is tempting, but that is the faster respiratory response, not a renal one.

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